

Aarav Pant

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EDUCATION

McMaster University

Hamilton, ON, Canada

Bachelor of Engineering in Computer Science (Honours, Co-op), **4th Year**

Sep 2022 – Apr 2027

Dean's Honour List, **GPA: 3.7/4.00**

- **Career Summary:** Fourth-year Computer Science student pursuing a career in **AI/ML Engineering and Cloud Security**, with experience spanning **16 months** of **Software Engineering** at Evertz, **cloud security tooling**, and **8 months** as a **Teaching Assistant** at McMaster. Targeting **CompTIA Security+** (Dec 2026), **AWS SAA-C03** (Jan 2027), and **Google Professional ML Engineer** (Feb 2027).
- **Certifications:** **CompTIA Security+ (SY0-701)** (*In Progress — Target: Dec 2026*), **AWS SAA-C03** (*In Progress — Target: Jan 2027*), Programming with JavaScript by Meta, Introduction to Front-End Development by Meta, Python for Data Science, AI & Development by IBM, Version Control by Meta.
- **Relevant Coursework:** Discrete Math, Introduction to Programming, Intro to Software Using Web Programming, Automata and Computability, Data Structures & Algorithms, Databases, Computer Architecture, Intro to Software Development, Concurrent Systems, Algorithms and Software Design, Algorithms and Complexity, Data Privacy, Principles of Programming Languages, Computer Networks and Security, Operating Systems, Syntax-Based Tools & Compilers, Applied Cryptography, Software Testing, Fundamentals of Machine Learning, Linear Optimization, Continuous Optimization, Capstone Project.

SKILLS

Programming Languages: Python, Java, C, C++, C#, Kotlin, JavaScript, TypeScript, HTML, CSS, Haskell, Elm

Databases: MySQL, MongoDB, PostgreSQL, SQLite

Frameworks, Libraries, and Tools: React, React Native, Next.js, Redux, Tailwind CSS, Bootstrap, Express, Django, Three.js, PyTorch, TensorFlow, Keras, Hugging Face Transformers, LangChain, NumPy, Scikit-learn, Firebase, FastAPI, Spring Boot, Angular, Matplotlib, Pandas, Docker, Kubernetes, Jenkins, Postman, JUnit, Git, GitHub, Jira, UML, Agile

Cloud and DevOps: AWS (S3, EC2, Lambda, IAM, RDS, DynamoDB, SNS, EventBridge, CloudWatch), GCP (Cloud Run), Terraform, GitHub Actions, Docker Compose, CI/CD Pipelines

EXPERIENCE

McMaster University

Hamilton, Ontario, Canada

Teaching Assistant, *COMPSCI 2XC3 & CS 1MD3*

Jan 2025 – Apr 2025, Sept – Dec 2026

- Supported students in **CS 1MD3** on programming fundamentals (control flow, recursion, exceptions, file I/O) and in **COMPSCI 2XC3** on **algorithm design & analysis** and **software design principles** through tutorials and guidance.
- Assisted with grading, technical documentation, and presentations, providing clear feedback to help students improve their problem-solving and coding skills.
- Invigilated **midterm and final examinations**, helping ensure a smooth, organized, and fair testing environment.

Evertz Microsystems Ltd.

Burlington, ON, Canada

Software Engineer Intern

May 2025 – August 2026

- Developed and executed comprehensive test cases for the **Mediator-X** platform, ensuring robust **media asset management** and playout functionalities across diverse client environments.
- Collaborated directly with clients to gather requirements and deliver customized solutions, including dynamic reports, **SQL queries**, and automation scripts, enhancing operational efficiency.
- Engineered automation tools using **Python** and **JavaScript** to streamline repetitive tasks within Mediator-X workflows, reducing manual intervention and improving system reliability.

Google Developer Student Club (GDSC) | [GitHub](#)

McMaster University, Hamilton, ON

Open Source Team Member (Software Developer)

Oct 2024 – April 2025

- **Collaborated with 9 developers** to design and implement **3 key features** for the DBAC Companion App using **React Native** and **Tailwind CSS**, enhancing usability for over 50 students.
- Redesigned the application interface, leading to a **10% increase in user satisfaction** and streamlining navigation workflows for a seamless user experience.
- **Utilized Google Cloud Platform (GCP)** for database management and **Firestore** for real-time data synchronization, ensuring efficient and scalable data handling.

PROJECTS

AWS Cloud Security Posture Scanner | *Python, boto3, Terraform, AWS, Kubernetes* | [GitHub](#)

- **Built an automated auditing tool** implementing **5 CIS AWS** and **4 CIS Kubernetes Benchmark checks**, validated live against a real AWS account and Kubernetes cluster.
- **Designed a serverless scan pipeline** (**Lambda + EventBridge + DynamoDB + SNS**) via **Terraform** with a findings lifecycle and **opt-in auto-remediation** for low-risk findings.
- **Built a CloudWatch dashboard** and custom metrics pipeline for scan latency and findings trends, backed by a least-privilege IAM role scoped per capability rather than one broad policy.

Security-Focused RAG Assistant | *FastAPI, PyTorch, Vector DB (Chroma), GCP Cloud Run, Docker* | [GitHub](#) | [Live Demo](#)

- **Built a RAG pipeline** over security/compliance docs with GPU-accelerated embeddings and a locally-hosted LLM, generating citation-grounded answers validated against a real NIST corpus.
- **Implemented a structural hallucination guard** that rejects generation when retrieval confidence falls below an empirically-derived threshold, rather than relying on prompt instructions alone.
- **Built a 28-question evaluation harness** measuring retrieval precision, citation accuracy, and keyword recall (**66.7% each**) with a **0% hallucination rate**, fixing 3 real bugs found along the way.

Job Application Tracker | *Next.js, TypeScript, Prisma, Tailwind CSS* | [GitHub](#)

- **Built a full-stack personal CRM** to track job/co-op applications, including a status pipeline (Applied → OA → Interview → Offer/Rejected) and a dashboard with real-time funnel conversion stats.
- **Developed a Prisma-backed Next.js application** with a follow-up reminder system that flags stale applications after 14 days.
- **Validated all API inputs at the boundary** using **Zod** schemas and indexed database queries to match actual access patterns.